

WASP-84b

R-1096

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-84_180220 · created 2026-07-30T18:28:30+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458169.7951 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.161 +/- 0.048
Transit depth (Rp/Rs)^2	2.58 +/- 1.55 [%]
Orbital Inclination (inc)	89.91 +/- 1.96
Airmass coefficient 1 (a1)	1.57 +/- 1.05
Airmass coefficient 2 (a2)	-0.53 +/- 0.68
Scatter in the residuals of the lightcurve fit is	70.72 %
Best Comparison Star	None
Optimal Aperture	43.43
Optimal Annulus	90.38
Transit Duration (day)	0.109 +/- 0.046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.161 ± 0.048 vs 0.1295 (deviation 0.48σ)
Duration fit vs published	0.109 d vs 0.1146 d (deviation 0.09σ)
Mid-time O–C	-5.9 min
Depth SNR	2.4
Residual scatter	70.72 %

Light curve

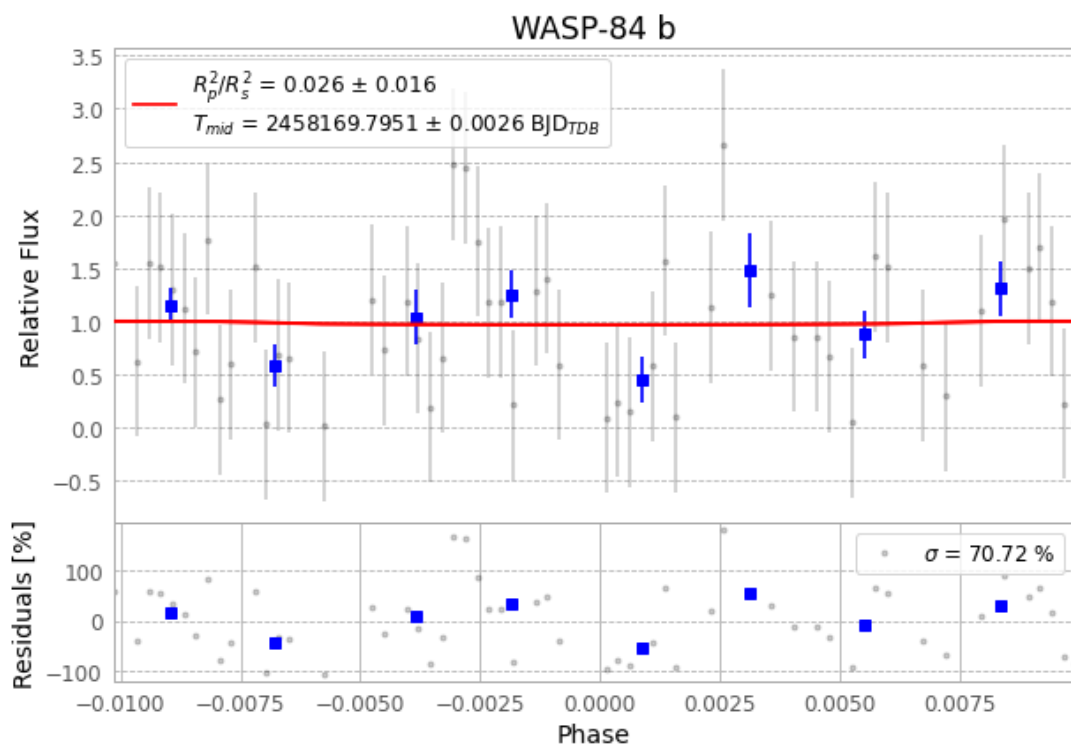


Figure 1 — FinalLightCurve_WASP-84 b_2018-02-19.png (SHA-256 20c25f28a7aa6637...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [166, 428], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4f6e118819f1ec3b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), WASP-84180220045110.FITS → WASP-84180220090012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-84-180220.log (c1b1dd21025f...), FinalLightCurve_WASP-84 b_2018-02-19.png (20c25f28a7aa...), FinalLightCurve_WASP-84 b_2018-02-19.csv (e35929c57aea...)

Compute resources

Wall time 355.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 18:28Z.